VIDURANGI KALPANA

Full-Stack Software Engineer | AI Engineer | Data Engineer | Data Scientist

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SUMMARY

Data Science and AI professional with **3 years of industry experience**, including 2.5 years in a US-based NLU & ML team, building production AI/ML systems that turn large volumes of unstructured data into business insight. Hands-on across the full engineering lifecycle with a proven record of delivering **advanced analytics use cases end-to-end** including conversation analytics, RAG-based search, and LLM evaluation. Strong in problem solving skills backed by a **BSc (Hons) in Computer Engineering** and an ongoing **MSc in Data Science & AI**. Proactive team member with experience translating technical findings into clear recommendations for product and client-facing stakeholders, with a proven ability to learn quickly and adapt to new environments. Looking for a long-term role in a collaborative, product-driven team, bringing hands-on expertise while continuing to grow with modern technologies.

PROFESSIONAL EXPERIENCE

Rozie AI Inc. — Durham, NC, USA (Remote from Sri Lanka)

Software Engineer II, NLU & ML Team *Jan 2026 - June 2026*

- Developed **analytics workflows, ETL pipelines and OLAP/OLTP data models** for an AI-powered conversation analytics platform, enabling NLP-driven, query-based insight generation over large volumes of customer conversation data.
- Designed and developed a **clustering-quality evaluation framework** for conversation transcripts - automating data ingestion, PII redaction, LLM-generated ground-truth extraction, semantic-equivalence scoring, noise-appropriateness checks and evaluation reporting to systematically validate model output.
- Integrated backend analytics APIs with a cloud-based conversation-insights dashboard, putting model outputs in front of business users as actionable insights.

Associate Software Engineer, NLU & ML Team *Jan 2024 - Jan 2026*

- Improved the accuracy and relevance of a **RAG-based question-answering search service** by enhancing semantic search with vector embeddings, transformer models and hybrid retrieval.
- Designed and developed an **end-to-end evaluation and testing framework for RAG/LLM applications**, strengthening model-quality assessment, observability and maintainability.
- Contributed to the architecture and end-to-end implementation of an AI-powered customer summary reporting capability - request validation, metadata management, cloud file storage and retrieval, LLM integration, prompt management, data aggregation and analytics.
- Developed data adapters for a data-connector framework, integrating third-party platforms to improve data ingestion and processing workflows.

Clouda Inc. (Pvt) Ltd — Irvine, CA, USA (Remote from Sri Lanka)

Intern Software Engineer *Dec 2022 - May 2023*

- Developed REST CRUD services and dashboard features using Go, TypeScript, Next.js, Docker & AWS for SaaS products.
- Researched API management, CDNs and serverless technologies to support cloud-native architecture decisions.

ACADEMIC EXPERIENCE

Data Science and AI Researcher *Jan 2023 - Present*

- Researched about "A Cost-Effective Intelligent Tutor using Large Language Models" and **presented as first author** at the **11th IEEE ICEEET 2026** (**~25%** acceptance rate), attended by researchers from 13 countries.
- Conducting ongoing research on privacy-preserving conversational AI using Reinforcement Learning.

Casual Instructor, Department of Computer Engineering, University of Peradeniya *Sept 2022 - Nov 2023*

- Conducted laboratory sessions for undergraduate modules: Network and Web Application Design (CO324), Data Structures and Algorithms (CO322), and Computer and Network Security (CO325).

TECHNICAL SKILLS

Programming	Python, Java, TypeScript, JavaScript, C, SQL
Generative AI	LLMs, Retrieval-Augmented Generation (RAG), hybrid retrieval, LangChain, LangGraph, PydanticAI, Amazon Bedrock
Machine Learning	Scikit-learn, PyTorch, TensorFlow, Amazon SageMaker, NLP, Clustering, Transformer Models, Vector Embeddings, Model Evaluation & Validation
Big Data & Engineering	Databricks, Apache Spark, Apache Airflow, dbt, ETL pipelines, OLAP/OLTP data modelling
Databases	PostgreSQL, MySQL, DuckDB, MongoDB, DynamoDB
MLOps & Cloud	AWS, GCP, Azure, Docker, Terraform, Git; LLM evaluation & observability (RAGAS, DeepEval, Langfuse)
Web & Collaboration	FastAPI, REST APIs, React.js, Angular, Node.js; Jira, Confluence

SELECTED PROJECTS

1. AI-Powered Conversation Analytics Platform — *Professional - Rozie AI*

Platform that processes customer conversation data to generate summaries, surface trends and support data-driven reporting through AI-assisted analytics workflows.

- **Role:** Backend development, ETL pipeline development, OLAP/OLTP data modelling, Analytics workflow development, Clustering-quality evaluation, API design & development, Dashboard integration.
- **Stack:** Python, FastAPI, LangGraph, LangChain, PostgreSQL, DuckDB, AWS (Bedrock, SageMaker, DynamoDB, S3, Cognito), Angular, TypeScript

2. RAG-Based Search & LLM Evaluation System *Professional - Rozie AI*

AI search system improving how LLM applications retrieve information and generate answers, with continuous monitoring and quality assessment of responses.

- **Role:** Semantic search enhancement, Data-connector development & integration, Design and development of the evaluation & testing framework.
- **Stack:** Python, LangChain, transformer models, vector embeddings, hybrid retrieval, Langfuse, DeepEval, RAGAS, Airbyte

3. ModuMate – Cost-Effective Intelligent Tutor Using LLMs *Research - Academic (UoP)*

Researched the challenges of LLM-based education platforms, focusing on the cost of LLM API access, and built a prototype to analyse and demonstrate cost reduction.

- **Role:** Prototype full-stack development, backend testing, Research and paper drafting
- **Stack:** LLM APIs, web application (frontend & backend) - Python, Angular, Flask, Docker

4. CareFlow – Medical Clinic Booking & Queue Management System *Freelance - Individual*

Built a full-stack clinic web app with account-free online booking, SMS reminders and an AI chat assistant for patients. Developed a role-based staff portal for scheduling, QR check-in, walk-ins and live queues, built to keep patient data private and secure.

- **Role:** Requirements gathering, system design, data modelling, backend REST API development, React frontend, AI chatbot integration, SMS notification pipeline, automated testing, containerisation, security hardening & production deployment.
- **Stack:** Python, FastAPI, PostgreSQL, React, TypeScript, Docker, AWS (CDK, ECS Fargate, RDS, S3, CloudFront, Bedrock, SNS, WAF, Route 53, CloudWatch), Pytest

5. ATM Cash Optimizer – ATM Cash Optimization & Demand Forecasting System *RnD - Individual*

End-to-end cash planning system for bank ATMs & CRMs that combines quantile demand forecasting with mixed-integer optimization to schedule cost-efficient refills. In backtests on real ATM data, it cut cash-handling costs by 35–42% compared with standard bank refill rules.

- **Role:** System design, data ingestion and validation pipeline, EDA, scenario data simulation, Feature engineering, Quantile forecasting and calibration, MILP optimization modelling, backtesting and simulation engine, DuckDB data warehousing, Streamlit dashboard, automated testing, CI/CD and project web page deployment.
- **Stack:** Python, PySpark, LightGBM, PuLP (MILP), DuckDB, SQL, MLflow, Streamlit, Plotly

EDUCATION

University of Moratuwa *Jan 2026 - Present*

MSc in Data Science & Artificial Intelligence - Department of Computer Science & Engineering

University of Peradeniya *Nov 2018 - Dec 2023*

BSc (Hons) Engineering in Computer Engineering - **Second Class Honours (Upper Division)**

Institute of Java and Software Engineering (IJSE) *Sep 2017 - Apr 2018*

Graduate Diploma in Software Engineering

PUBLICATIONS

- **M. W. V. Kalpana**, V. Navanjana, T. Manohara, I. Nawinna, R. Ragel and D. Herath, "*A Cost-Effective Intelligent Tutor using Large Language Models*" 2026 IEEE International Conference on Electrical Engineering and Emerging Technologies (ICEEET), Kandy, Sri Lanka, Oct. 2026.

CERTIFICATIONS

- **Academy Accreditation - Databricks Fundamentals** | Databricks *2026*
- **Data Engineering Professional Certificate** | DeepLearning.AI & AWS (Coursera) *2026*
- **Fundamentals of Accelerated Data Science** | NVIDIA *2026*
- **AWS Academy Graduate - Machine Learning for NLP** | AWS *2026*
- **AWS Academy Graduate - Data Engineering** | AWS *2026*
- **Generative AI Leader Certification** | Google Cloud *2026*
- **AWS Certified AI Practitioner** | AWS *2025*
- **Neural Networks and Deep Learning** | DeepLearning.AI (Coursera) *2022*
- **Complete Machine Learning & Data Science Bootcamp** | Udemy *2022*

AWARDS & LEADERSHIP

- **First Place** in HACKFEST, University of Peradeniya (2022) and **First Runner-Up** in Code With WIE, IEEE WIE Sri Lanka (2022) - Smart Technological solution of Sustainable Agriculture for the Project Greenhouse Monitoring & Control System Based on Image Spectral Data and IoT Sensor Data
- **Captain / Vice Captain**, University of Peradeniya Women's Badminton Team (2021 - 2023), **Full Colours Holder and Lead Player** (2019 - 2023)
- **Vice President - Finance**, Organizing Committee, Unlocked 2.0 - AIESEC in Kandy (2021)
- **National-level School Badminton Player** with achievements at zonal, provincial and all-island levels (2007 - 2016)
- **Senior Prefect/ Assistant Prefect** at G/Dharmasoka College (2014 - 2016)